

ETHENA · SECURITY REVIEW

Ethena

PSMAdapter Review.

A security review of PSMAdapter Review.

CLIENT	PROJECT	AUDITORS	AUDIT WINDOW	FINAL REPORT
Ethena	PSMAdapter Review	Guardian	August 5 – August 10, 2026	August 17, 2026

§ 00

Contents.

17 pages — overview, confidence ranking, scope, 8 findings, disclaimer, and about Guardian.

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§ 01

Overview.

Guardian reviewed PSMAdapter Review across 2 review rounds. The final report includes 1 Low, 7 Informational findings.

AUDIT FIRM

Guardian

CLIENT

Ethena

PROJECT

PSMAdapter Review

PREPARED BY

Guardian

AUDIT WINDOW

August 5 – August 10, 2026

FINAL REPORT

August 17, 2026

METHODOLOGY

Smart contract · Manual · Main Review ·
Remediation Review

1.1 Confidence Ranking

5 / 5

Very High
Confidence

Guardian assigns a Confidence Ranking of 5 out of 5, combining finding severity, remediation outcome, and the reviewer’s assessment of latent risk. The full scale is on the next page.



1.2 Findings at a Glance

SEVERITY	TOTAL	PENDING	ACK	PARTIAL	RESOLVED
Critical	0	0	0	0	0
High	0	0	0	0	0
Medium	0	0	0	0	0
Low	1	0	1	0	0
Informational	7	0	1	0	6
All severities	8	0	2	0	6

§ 02

Confidence Ranking.

Guardian assigns a project-level confidence score using the scale below. The ranking combines finding severity, remediation outcome, and the reviewer's assessment of latent risk.

CONFIDENCE RANKING	DEFINITION AND RECOMMENDATION	RISK PROFILE
5 Very High Confidence ASSIGNED	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: code is highly secure at time of audit.	0 High/Critical findings and few Low/Medium severity findings.
4 High Confidence	Code is clean, well-structured, and adheres to best practices. Design patterns are sound and test coverage is strong. Recommendation: suitable for deployment after remediations; consider periodic review with changes.	0-1 High/Critical findings per engagement week and little to no Medium severity findings.
3 Moderate Confidence	Code is functional but shows gaps in defensive programming or test coverage. Recommendation: remediate the findings and re-review the changes before deployment.	Some High/Critical findings, with various Medium severity findings.
2 Low Confidence	Recurring issues suggest systemic gaps rather than isolated mistakes. Recommendation: remediate, then commission a follow-up review at a frozen commit.	Over 6 High and Critical findings across the engagement.
1 Very Low Confidence	The volume and severity of findings indicate the codebase is not ready for production. Recommendation: substantial rework, then a full re-review.	Over 12 High and Critical findings across the engagement.

§ 03

Scope & Classifications.

Guardian reviewed PSMAdapter Review.

3.1 Severity Matrix

	IMP · HIGH	IMP · MED	IMP · LOW
LIK · HIGH	Critical	High	Medium
LIK · MED	High	Medium	Low
LIK · LOW	Medium	Low	Low

IMPACT · HIGH

Significant loss of assets, harm to a group of users, or core protocol functionality is disrupted.

IMPACT · MEDIUM

Small amount of funds can be lost, or ancillary functionality is affected. Users may experience reduced or delayed receipt.

IMPACT · LOW

Unexpected behavior in some functionality that is notable but does not meet the criteria for higher severity.

LIKELIHOOD · HIGH

Possible with reasonable assumptions that mimic on-chain conditions; cost is low compared to amount gained.

LIKELIHOOD · MEDIUM

Possible only in uncommon cases or requires a large amount of capital relative to disruption caused.

LIKELIHOOD · LOW

Unlikely to ever occur in production.

Files in Scope.

FILE	NSLOC	TOTAL
No files were recorded in scope.		

§ 04

Findings.

Grouped by round. Visible IDs reset by severity within each round, so a finding is named the same way in the report as it was in the channel it was written in.

ID	TITLE	CATEGORY	SEVERITY	STATUS	PAGE
MAIN REVIEW					
L-01	Stale sell liquidity view	Unexpected Behavior	LOW	ACKNOWLEDGED	p. 8
I-01	Adapter uses generic low amount error	Informational	INFORMATIONAL	ACKNOWLEDGED	p. 9
I-02	Underdocumented same beneficiary reason	Documentation	INFORMATIONAL	RESOLVED	p. 10
I-03	Adapter lacks per-investor authorization	Informational	INFORMATIONAL	RESOLVED	p. 11
I-04	availableAsset() may show dust capacity	Informational	INFORMATIONAL	RESOLVED	p. 12
I-05	PSM rotation can move pending fees	Informational	INFORMATIONAL	RESOLVED	p. 13
I-06	Incomplete IGroveBasinLike documentation	Documentation	INFORMATIONAL	RESOLVED	p. 14
REMEDIATION REVIEW					
I-01	README misstates minimum input units	Documentation	INFORMATIONAL	RESOLVED	p. 15

LOW **L-01**

ACKNOWLEDGED

Stale sell liquidity view

CATEGORY	Unexpected Behavior
LOCATION	https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/IPSMAdapter.sol#L109-L110
REVIEW	Main Review

DESCRIPTION

The adapter integration documents that Securitize must update

`ExternalAssetProvider.availableAsset()` to delegate to `EthenaPSMAdapter.availableAsset()`, but there is no equivalent sell-side liquidity view for `ExternalLiquidityProvider.availableLiquidity()`. In the Securitize ELP implementation, `availableLiquidity()` resolves the liquidity custodian through `getLiquidityCustodian() / _custodianOf()` and, when the external provider is `EthenaPSMAdapter`, this reports `USDC.balanceOf(address(adapter))`.

That value does not represent redeem-side executable liquidity. Sell output is sourced by `PSM.swap()` from the PSM collateral send custodian and is bounded by custodian balance, custodian allowance, and sell-direction PSM caps. As a result, `availableLiquidity()` can underreport liquidity when the adapter holds no USDC, or overreport liquidity when USDC fees temporarily sit on the adapter before `sweep()`.

This does not directly block redemptions because `ExternalLiquidityProvider.supplyExactIn()` does not use `availableLiquidity()` as an execution precheck, but frontends or off-chain integrators using the public off-ramp liquidity view may display or act on stale liquidity data.

RECOMMENDATION

Consider adding a sell-direction liquidity view to `EthenaPSMAdapter` that accounts for collateral send custodian balance/allowance and `swapForCollateral` PSM caps, and have Securitize delegate `ExternalLiquidityProvider.availableLiquidity()` to it. Alternatively, explicitly document that `availableLiquidity()` is not meaningful for this adapter integration and should not be used for off-ramp capacity decisions.

RESOLUTION

Status: Acknowledged.

INFORMATIONAL **I-01**

ACKNOWLEDGED

Adapter uses generic low amount error

CATEGORY	Informational
LOCATION	https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/EthenaPSMAdapter.sol#L165 , https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/EthenaPSMAdapter.sol#L195
REVIEW	Main Review

DESCRIPTION

In `swapExactIn()`, the adapter checks `amountIn < PSM_MIN_AMOUNT_IN` before resolving the swap direction with `_resolveDirection()`. As a result, both collateral-to-asset and asset-to-collateral swaps revert with the same `AmountInBelowPsmMinimum` error, instead of preserving the PSM's direction-specific distinction between `InvalidCollateralAmount` and `InvalidAssetAmount`.

RECOMMENDATION

Consider whether the adapter should keep the current generic `AmountInBelowPsmMinimum` error for simplicity, or introduce direction-specific low-amount errors.

RESOLUTION

Status: Acknowledged.

INFORMATIONAL I-02

RESOLVED

Underdocumented same beneficiary reason

CATEGORY	Documentation
LOCATION	README.md
REVIEW	Main Review

DESCRIPTION

The `README` documents the following key invariant.

> Key invariant: the adapter always sets beneficiary: address(this) so it can measure the balance delta (received == amountOut) before forwarding funds to the ramp.

While this is technically true, there is another reason why `beneficiary` must be set to the adapter and it is because the benefactor of the order is the adapter as well.

`PSM._validateBenefactor()` reverts if the benefactor and the beneficiary differ, unless the beneficiary is approved for the given benefactor.

```
if (order.benefactor != order.beneficiary &&
!_benefactorState.config.approvedBeneficiaries[order.beneficiary])
{
    revert BeneficiaryNotApproved(order.beneficiary);
}
```

Because the adapter has no way to call `setApprovedBeneficiary()`, this check would have always reverted if the two didn't match.

RECOMMENDATION

Add this requirement to the key invariant documentation.

RESOLUTION

Status: Resolved.

INFORMATIONAL I-03

RESOLVED

Adapter lacks per-investor authorization

CATEGORY	Informational
LOCATION	https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/EthenaPSMAdapter.sol#L234-L235
REVIEW	Main Review

DESCRIPTION

`EthenaPSMAdapter.swapExactIn()` does not identify or authorize the underlying investor. On the Securitize public on-ramp path, investor access is gated by the `BUIDL` /Securitize registry and token compliance checks rather than a ramp-specific allowlist. Therefore, any token-level eligible investor can buy or sell through the configured ramp path. The downstream `PSM` order sets both `benefactor` and `beneficiary` to `address(this)`, so all ramp users consume the same adapter-level benefactor caps rather than separate per-investor caps.

RECOMMENDATION

Consider documenting this trust boundary explicitly. If shared token-level investor eligibility is intended, no code change is required. If Ethena wants granular per-investor or per-ramp access control, add authorization or limits in the Securitize ramp/provider path.

RESOLUTION

Status: Resolved.

INFORMATIONAL **I-04**

RESOLVED

availableAsset() may show dust capacity

CATEGORY Informational

LOCATION <https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/EthenaPSMAdapter.sol#L303>

REVIEW Main Review

DESCRIPTION

`EthenaPSMAdapter.availableAsset()` returns the remaining `BUIDL` capacity without considering `PSM_MIN_AMOUNT_IN`. If the amount of `BUIDL` can be received by paying USDC below the floor, `availableAsset()` will return that `BUIDL` amount, even though it cannot be executed.

This inconsistency affects only `view` paths, since `EAP.supplyExactIn()` calls `previewSwapExactIn()` before `availableAsset()`, and `previewSwapExactIn()` reverts if the provided amount is below the floor.

RECOMMENDATION

If this is fine, acknowledge the finding. If not, return 0 in case the `available` variable is below the minimum executable amount.

```
uint256 minExecutableOut = _pegAmountOut(uint128(PSM_MIN_AMOUNT_IN), true);  
if (available < minExecutableOut) return 0;
```

RESOLUTION

Status: Resolved.

INFORMATIONAL I-05

RESOLVED

PSM rotation can move pending fees

CATEGORY	Informational
LOCATION	https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/EthenaPSMAdapter.sol#L313-L319 , https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/EthenaPSMAdapter.sol#L356-L366
REVIEW	Main Review

DESCRIPTION

`EthenaPSMAdapter.setPsm()` rotates the configured `PSM` without first sweeping token balances held by the adapter. `EthenaPSMAdapter.sweep()` always reads the current `PSM` and sends pending USDC/BUIDL balances to the current PSM custodian addresses.

As a result, if USDC ramp fees or stray BUIDL are present on the adapter when `setPsm()` is called, a later `sweep()` sends those pre-rotation balances to the new PSM custodians rather than the custodians configured on the previous PSM.

RECOMMENDATION

Document that operators should call `sweep()` before `setPsm()` when balances should be attributed to the previous PSM custodian set.

RESOLUTION

Status: Resolved.

INFORMATIONAL I-06

RESOLVED

Incomplete IGroveBasinLike documentation

CATEGORY	Documentation
LOCATION	https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/IGroveBasinLike.sol#L85-L89 , https://github.com/GuardianOrg/onchain-minting-internal-team1-1785887214892/blob/d015d3b0a3a2bf401135bf9267a07c44602e5597/src/basin/IGroveBasinLike.sol#L101
REVIEW	Main Review

DESCRIPTION

The `previewSwapExactIn()` documentation states that the preview may overestimate when the PSM oracle price deviates below `pegPrice`. This only describes the buy-direction case.

Since `previewSwapExactIn()` replicates only the peg path while `swapExactIn()` uses `psm.getQuote()` and returns the lower of the peg path and oracle path, overestimation for sells can happen whenever the oracle path is lower than the peg path. Whether this occurs below or above `pegPrice` depends on direction: below `pegPrice` for buys, and above `pegPrice` for sells.

Additionally, the `previewSwapExactIn` and `swapExactIn()` documentations refers to `tokenIn` and `tokenOut` while the actual parameters are named `assetIn` and `assetOut`.

RECOMMENDATION

Consider updating the `previewSwapExactIn()` documentation to state that overestimation can occur whenever the oracle path wins, with the side of peg depending on swap direction. Also replace `tokenIn/tokenOut` with `assetIn/assetOut`.

RESOLUTION

Status: Resolved.

INFORMATIONAL I-01

RESOLVED

README misstates minimum input units

CATEGORY	Documentation
LOCATION	https://github.com/GuardianOrg/onchain-minting-internal/blob/c6e5678552635b215adec6c17c3a340f3860434a/src/basin/README.md#L75-L76
REVIEW	Remediation Review

DESCRIPTION

The README states that amounts below `PSM_MIN_AMOUNT_IN` revert in collateral-token units. However, `EthenaPSMAdapter.swapExactIn()` applies `PSM_MIN_AMOUNT_IN` to `amountIn` before resolving the swap direction, so the floor is denominated in the input token's units. For sells, this means the threshold is in asset decimals, not collateral decimals.

RECOMMENDATION

Update the README to state that `PSM_MIN_AMOUNT_IN` is enforced in input-token units before direction resolution.

RESOLUTION

Status: Resolved.

§ 05

Disclaimer.

This report is not, nor should be considered, an "endorsement" or "disapproval" of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any "product" or "asset" created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian's position is that each company and individual are responsible for their own due diligence and continuous security. Guardian's goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis.

Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract's safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

§ 06

About Guardian.

Guardian is a smart contract security firm. Every engagement is worked by two independent teams competing on the same scope, each with its own tooling and methods, so no single reviewer's blind spot survives.

\$45B+DIGITAL ASSETS UNDER
REVIEW**200+**

ENGAGEMENTS COMPLETED

500+CRITICAL & HIGH FINDINGS
REPORTED

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